

Supervised Learning Project

Problem statement

You are a junior video game developer working to improve the player experience in an MMORPG. You have deployed several seasonal NPCs throughout big cities in the game. They all have equivalent quests and trade items, but they vary in level of friendliness.

The friendlier NPCs will give the player access to quests and trade items immediately or with minimal conversation. The less friendly NPCs will give the player access to quests and trade items only after a conversation, where the player establishes trust and obtains knowledge about the lore of the game.

You are secretly conducting an experiment with these seasonal NPCs to understand the level of interest in the RPG aspect of the game, so your data engineering team has been collecting data on user interactions with the seasonal NPCs. You plan to use the knowledge you get from these seasonal/temporary NPCs to make your NPCs across the game more engaging in the next patch of the game.

Project Goal

Use any of the supervised learning methods discussed in class to decide what changes, if any you need to make to your nominal NPCs for the next patch (called old NPCs in the datasets).

Goal 1: Discover and quantify what drives user engagement. (in other words, describe engagement with an equation that can predict it). You will need to:

- Define what you mean by user engagement and how you will measure it.
- Explore the data to find out what features of the NPCs and/or interactions drive user engagement.
- Engineer new features to define user engagement and/or to help you find out what drives it.
- Use one of the learning models discussed in class to find out what drives user engagement and to predict it.

Goal 2: Make a recommendation for how NPCs could be improved in the game for the coming patch. You will need to:

Available Datasets

The **interactions** dataset contains the data that you will examine for this task. The **interactions** dataset has one user interaction per row. For reference, the stats of your current NPCs are found in the **oldNPCs** dataset and the stats for your experiment NPCs are found in the **seasonalNPCs** dataset.

Instructions

Use one of the supervised learning methods discussed in class to decide what changes, if any you need to make to your nominal NPCs for the next patch. You want to **optimize for user engagement with NPCs**. You will be graded on an **6-8 minute presentation** of your findings, including how you decided to measure user engagement. Your presentation must include:

1. Introduction to your game (extra credit only).
2. Visualization of the **old NPCs** and **seasonalNPCs** datasets. Answer these questions: What changed in the Seasonal (experimental) NPCs with respect to the old ones?
3. Visualization of the **interactions** dataset and definitions including:
 - Any features you engineered
 - How you defined user engagement
 - Any categories you defined or used to better understand the data
4. One slide explaining what method(s) you used to model engagement and why.

- Include models, training methods, and a list of target & feature(s).
- 5. Visualization of the training process (learning curve, weights, etc.)
 - Include the final model(s) (equations) you found that describe and can predict user engagement.
- 6. Results and recommendations for the next patch including:
 - Changes to existing NPCs (feedback to the game design team)
 - New data to collect (feedback for data engineering team)
- 7. Plan for how you will measure success of the suggested changes after the next patch including:
 - what metrics you will use
 - what algorithms you will employ

Rubric

Points	Presentation
10	Visualization NPC datasets + interpretation
10	Visualization interactions dataset + interpretation
10	Optimal methods chosen & correct answers
10	Visualization of the training process
10	Results and recommendations (flow from analysis)
10	Plan for success metrics
10	Correct application & interpretation of class concepts
10	Code (clearly commented)
10	Quality of slides (clear, visual, not text-heavy)
10	Delivery fluency + clarity
+10	AI-generated video game concept

Small things, when neglected, can bring down great endeavors.

Deduction	Criterion
-10	Under/Over time > 1 minute
-15	Poorly explained content/methods
-10	Missing axis labels, legends, etc. on any of your plots
-10	Poor handling of technology in the room
-10	Failing to give credit to classmates/AI tools when appropriate
-20	Missing PDF slides on Canvas by the start of the talk
-10	Missing colab notebook (link & PDF) on Canvas